

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 12-09-2026

1. TITLE

Design a College Information Webpage Using HTML

2. AIM

To create a basic college information webpage using HTML and demonstrate the use of headings, paragraphs, text formatting, lists, hyperlinks, images, tables, forms and multimedia elements.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
 - To organize webpage content using semantic HTML elements.
 - To demonstrate different types of lists and hyperlinks.
 - To display student information using an HTML table.
 - To create a registration form using different input controls.
 - To embed audio and video content into a webpage.
 - To use text-formatting elements such as bold, italic, underline, subscript and superscript.
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4. THEORY

HTML or HyperText Markup Language, is the standard markup language used to structure content on webpages. An HTML document consists of elements such as headings, paragraphs, sections, links, images, tables and forms. In this experiment, a college information portal is created using basic HTML elements. The page contains a header with navigation links, an information section, lists describing courses and facilities, useful external links, images, a student-details table and a registration form. The registration form demonstrates several HTML input types including text, email, password, date, number, color, range, radio buttons and checkboxes. A dropdown list and textarea are also included.

5. ALGORITHM / PROCEDURE

1. Create a new HTML document and define the basic HTML structure.
2. Add the required metadata and webpage title inside the <head> section.
3. Create a header containing the college portal heading. Add navigation links for the major sections of the webpage.
5. Create an introductory college-information section.
6. Demonstrate text formatting using bold, italic, underline, subscript and superscript.
7. Add ordered and unordered lists for academic programs and campus facilities.
8. Insert external hyperlinks using the anchor element.
9. Add college-related images using the element. Create a table containing sample student information.
10. Develop a student registration form using different HTML form controls.
11. Add audio and video elements with controls. Add a footer containing copyright information.
12. Save the file as code.html. Open the file in a browser and verify all webpage sections.

6. PROGRAM

code.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>College Information Portal</title>
</head>

<body>

    <!-- Header Section -->
    <header>
        <h1>College Information Portal</h1>

        <nav>
            <a href="#about">About</a> |
            <a href="#students">Students</a> |
            <a href="#registration">Registration</a>
        </nav>
        <hr>
    </header>

    <!-- About College -->
    <section id="about">

        <article>

            <h2>About Our College</h2>

            <p>
                Welcome to <b>ANITS Engineering College</b>.
                Our college provides
                <i>quality education</i> and encourages
                <u>innovation</u>.
            </p>

            <p>
                Water Formula: H<sub>2</sub>O <br>
            </p>
        </article>
    </section>
</body>
</html>
```

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        Einstein Equation:  $E = mc^2$ 
    </p>

</article>

</section>

<hr>

<!-- Lists -->
<section>

    <h2>Courses Offered (Ordered List)</h2>

    <ol>
        <li>Computer Science Engineering</li>
        <li>Electronics Engineering</li>
        <li>Mechanical Engineering</li>
        <li>Civil Engineering</li>
    </ol>

    <h2>Facilities (Unordered List)</h2>

    <ul>
        <li>Library</li>
        <li>Computer Labs</li>
        <li>Hostel</li>
        <li>Sports Ground</li>
    </ul>

</section>

<hr>

<!-- Hyperlinks -->
<section>

    <h2>Useful Links</h2>

    <a href="https://www.google.com" target="_blank">Google</a><br><br>

    <a href="https://www.yourcollege.edu" target="_blank">
        College Website
    </a>

</section>

<hr>

<!-- Images -->
<section>

    <h2>College Images</h2>

    <!-- Replace logo.png with your image -->
    

    <br><br>

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        <!-- Replace college.jpg with your image -->
        

</section>

<hr>

<!-- Student Table -->
<section id="students">

    <h2>Student Details</h2>

    <table border="1" cellpadding="10">

        <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Branch</th>
            <th>Email</th>
        </tr>

        <tr>
            <td>101</td>
            <td>Rahul</td>
            <td>CSE</td>
            <td>rahul@gmail.com</td>
        </tr>

    </table>

</section>

<hr>

<!-- Registration Form -->
<section id="registration">

    <h2>Student Registration Form</h2>

    <form>

        <label>Name:</label><br>
        <input type="text" required><br><br>

        <label>Email:</label><br>
        <input type="email" required><br><br>

        <label>Password:</label><br>
        <input type="password"><br><br>

        <label>Date of Birth:</label><br>
        <input type="date"><br><br>

        <label>Age:</label><br>
        <input type="number" min="16" max="30"><br><br>

        <label>Favorite Color:</label><br>

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        <input type="color"><br><br>

        <label>Programming Skill:</label><br>
        <input type="range" min="0" max="100"><br><br>

        <label>Gender:</label><br>

        <input type="radio" name="gender"> Male
        <input type="radio" name="gender"> Female
        <input type="radio" name="gender"> Other

        <br><br>

        <label>Hobbies:</label><br>

        <input type="checkbox"> Reading
        <input type="checkbox"> Sports
        <input type="checkbox"> Music

        <br><br>

        <label>Branch:</label><br>

        <select>

            <option>CSE</option>
            <option>ECE</option>

        </select>

        <br><br>

        <label>Comments:</label><br>

        <textarea rows="5" cols="40"></textarea>

        <br><br>

        <input type="submit" value="Submit">

        <input type="reset" value="Reset">

    </form>

</section>

<hr>

<!-- Audio -->
<section>

    <h2>College Anthem</h2>

    <audio controls>
        <source src="anthem.mp3" type="audio/mpeg">
        Your browser does not support audio.
    </audio>

</section>

```

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<hr>

<!-- Video -->
<section>

    <h2>College Introduction Video</h2>

    <video width="500" controls>

        <source src="college.mp4" type="video/mp4">

        Your browser does not support video.

    </video>

</section>

<hr>

<!-- Footer -->
<footer>

    <p>
        © 2026 ANITS Engineering College | All Rights Reserved
    </p>

</footer>

</body>

</html>

```

7. CODE EXPLANATION

Code	Explanation
header / nav	Creates the page heading and navigation links.
<section> / <article>	Organizes the portal into logical content sections.
/	Displays ordered courses and unordered facilities.
	Creates links to external pages.
	Places the college logo and campus image using source files.
<table>	Displays student details in rows and columns.
<form>	Groups the student registration controls.
<input>	Provides different form controls such as text, date, number, color, range, radio and checkbox.
<select> / <textarea>	Provides branch selection and a comments field.
<audio> / <video>	Adds media controls to the page.
<footer>	Displays the final copyright information.

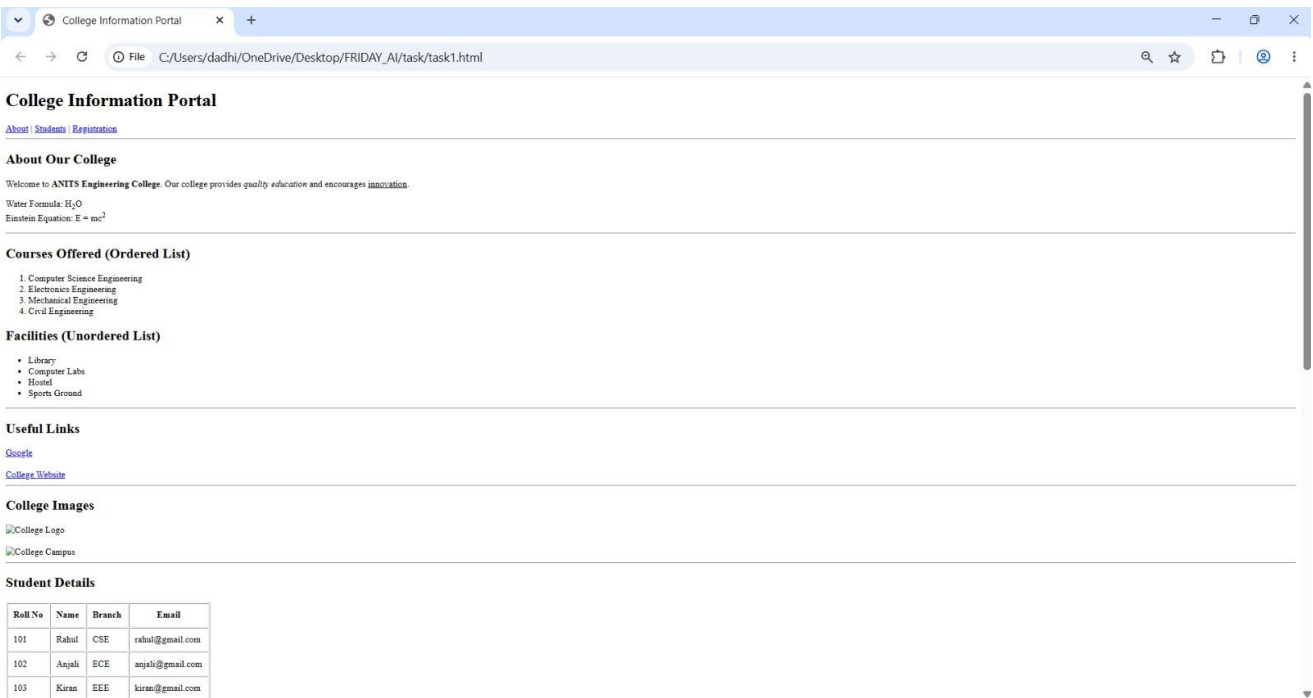
8. TEST CASES AND EXECUTION DATA

The following observations are based on the provided repository output screenshots. No unshown execution result has been invented.

Test Case	Input / Feature	Expected Result	Actual Result	Status
TC-01	Header, navigation and About section	Sections should be displayed at the top of the page	Header, navigation and About section are visible in output 1	Observed
TC-02	Courses, facilities, links, images and table	Corresponding elements should render	Lists and links render; image placeholders and student table are visible	Observed
TC-03	Registration form	All specified form controls should render	Registration controls are visible in output 2	Observed
TC-04	Audio and video elements	Media controls should be present	Audio control and video area are visible in output 2	Observed

9. OUTPUT

Output 1: Main portal and student details



Output 2: Student registration form along with audio and video elements.

The screenshot displays a web browser window titled 'College Information Portal'. The address bar shows the file path 'C:/Users/dadhi/OneDrive/Desktop/FRIDAY_AI/task/task1.html'. The page content is organized into three main sections:

- Student Registration Form:** This section contains several input fields and controls:
 - Name:
 - Email:
 - Password:
 - Date of Birth: (with a calendar icon)
 - Age:
 - Favorite Color:
 - Programming Skill:
 - Gender: ☐ Male ☐ Female ☐ Other
 - Hobbies: ☐ Reading ☐ Sports ☐ Music
 - Branch: (with a dropdown arrow)
 - Comments:
- College Anthem:** This section features an audio player with a play button, a progress bar at 0:00 / 0:00, and volume controls.
- College Introduction Video:** This section contains a video player, which is currently showing a black frame.

10. RESULT

Thus, a college information webpage was successfully developed using HTML. The webpage demonstrates structured content, text formatting, lists, hyperlinks, images, tabular data, form controls and multimedia elements. The implemented sections were verified using the provided webpage output.